

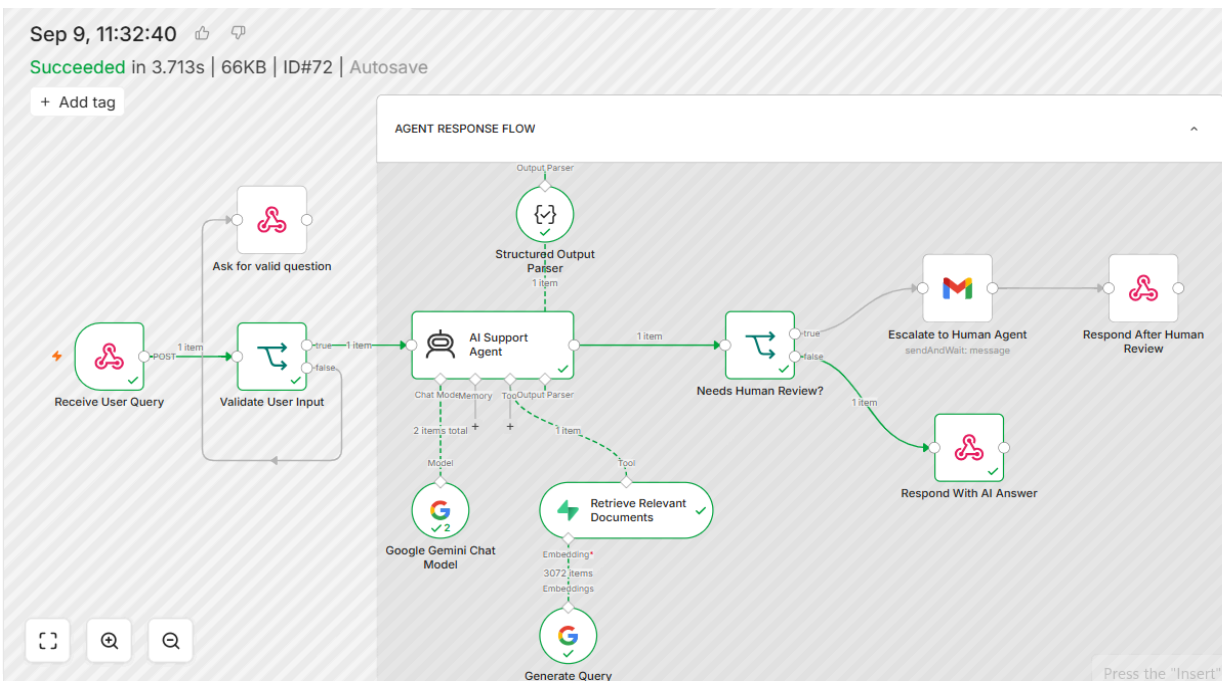
Case Study #2 — AI Customer Support RAG System (n8n + Google Gemini + Supabase)

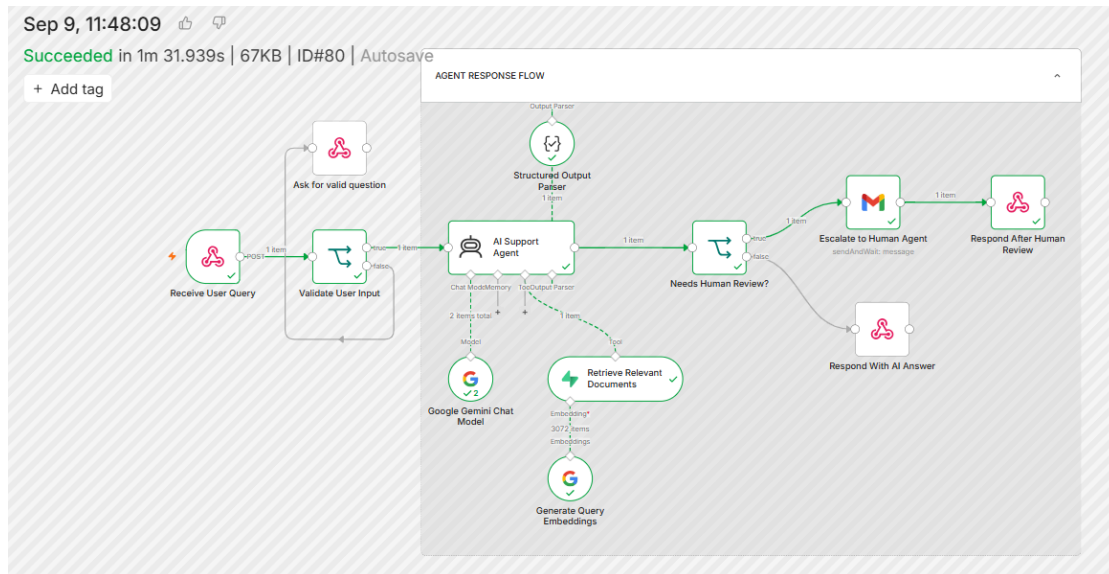
Project Summary

This project is an AI Customer Support RAG (Retrieval-Augmented Generation) system built with n8n.

It indexes internal support documentation, converts the content into embeddings, stores them in a Supabase-backed vector database, and answers customer questions using Google Gemini with retrieved knowledge-base context.

If the system detects missing information or low confidence, it automatically escalates the query for human review and responds after approval.





End-to-end support query workflow (validate → retrieve docs → AI answer → optional human escalation).

The Problem

Support teams often deal with repetitive questions and scattered documentation. This can cause:

- Slow response times
- Inconsistent answers
- Too much manual searching
- Difficult escalation of complex questions

Keeping support answers connected to the latest internal documentation can also be difficult.

What I Built

I built a two-part RAG automation system.

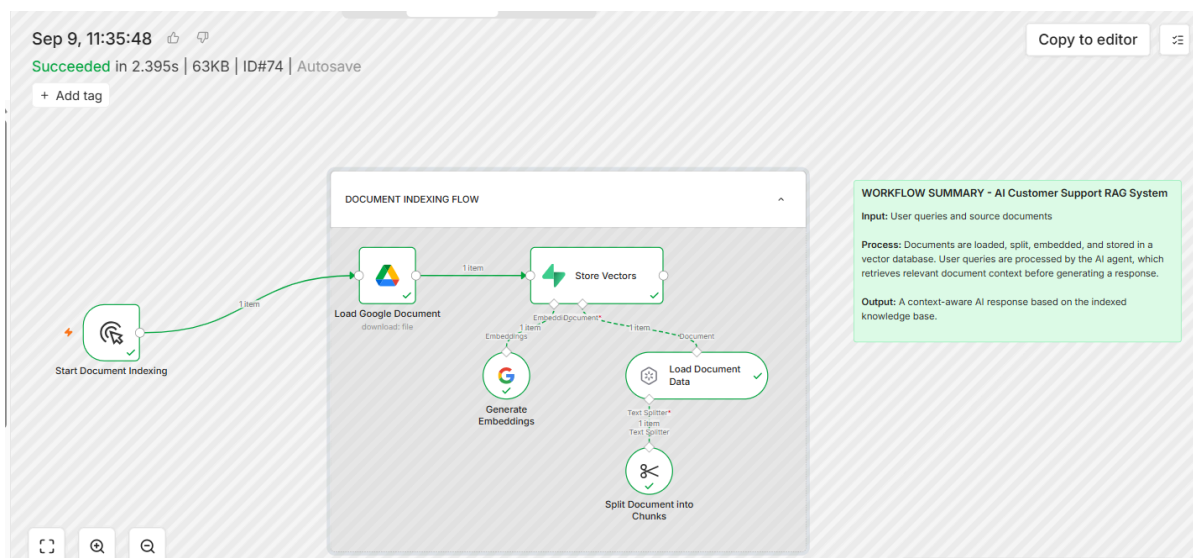
1) Document Indexing

Google Docs → Chunking → Gemini Embeddings → Supabase

The workflow:

- Loads support documentation
- Splits content into smaller chunks
- Creates embeddings
- Stores the vectors in Supabase

This creates a searchable knowledge base for the AI agent.



Document indexing pipeline (Docs → chunking → embeddings → Supabase vector storage).

2) AI Support Flow

When a user sends a question:

Webhook → Validate Input → Retrieve Knowledge → Gemini → Decision → Response

The AI agent:

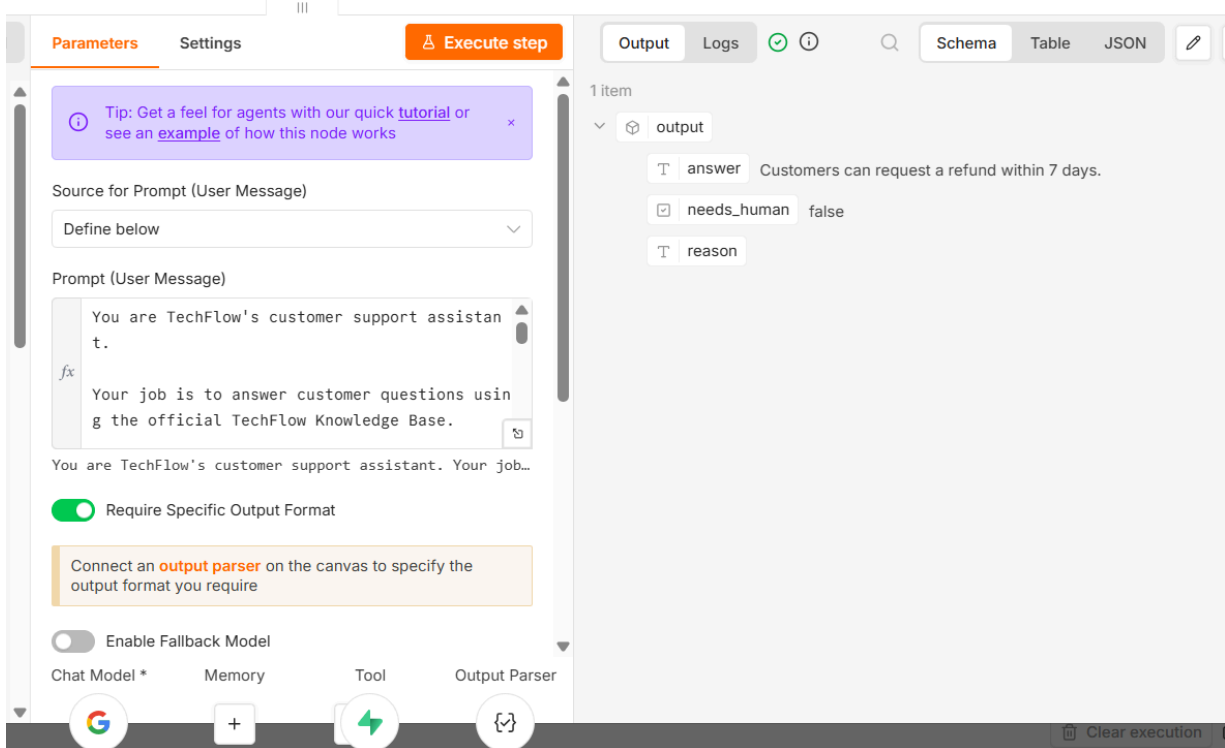
- Receives the question
- Searches the knowledge base
- Retrieves relevant information
- Generates an answer using that context

The screenshot displays the n8n workflow editor. On the left, the 'Parameters' tab is active, showing the configuration for a tool. A tip box at the top suggests using a RAG starter template. The 'Credential' is set to 'Supabase account'. The 'Operation Mode' is 'Retrieve Documents (As Tool for AI Agent)'. The 'Description' field contains a prompt: 'Use this tool to search the company's official knowledge base. Always search the knowledge base before answering customer questions.' The 'Table Name' is set to 'gemeniem'. An 'Execute step' button is visible at the top right of the configuration panel.

On the right, the 'Output' tab shows the result of the tool execution. It displays a single item with a 'response' field. The 'response[0]' field contains a JSON object with a 'pageContent' field. The 'pageContent' field contains the following text:

```
{"pageContent": "Company Name: TechFlow\n\nBusiness Hours:\n\nMonday-Friday: 9 AM - 6 PM\n\nServices:\n\nWe provide SEO, AI automation and web development services.\n\nPricing:\n\nSEO starts from $300/month.\n\nAI automation projects start from $500.\n\nRefund Policy:\n\nCustomers can request a refund within 7 days.\n\nSupport Email:\n\nsupport@techflow.com\n\nResponse Time:\n\nOur support team usually responds within 24 hours.\n\nPayment Methods:\n\nWe accept Visa, Mastercard and PayPal."}
```

RAG proof — relevant knowledge-base context is retrieved before generating the response



Gemini returns structured output (answer + needs_human flag + reason) for reliable routing.

RAG in Action

The important part of this workflow is that Gemini does not answer from the prompt alone.

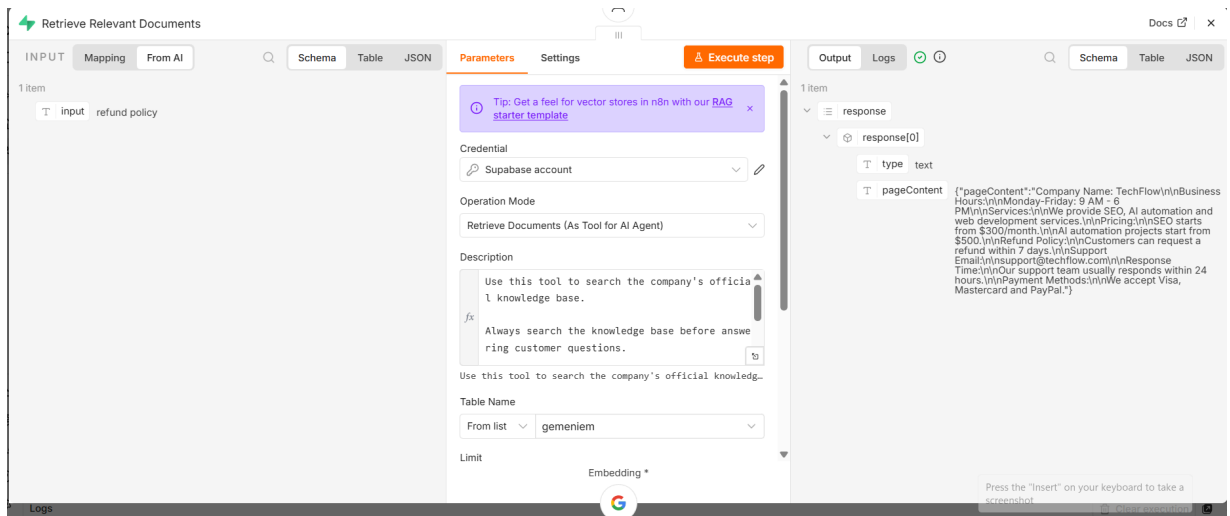
It first retrieves relevant knowledge-base content and uses that information to generate the response.

Output

```
{  
  "answer": "Your answer here",  
  "needs_human": false,  
}
```

```
"reason": "The knowledge base contains the required information."
}
```

This structured output makes the result easier to route automatically.



Human Review

When the knowledge base does not contain enough information, the workflow can escalate the question for human review.

Example

User Question

↓

Knowledge Retrieval

↓

Not Enough Information

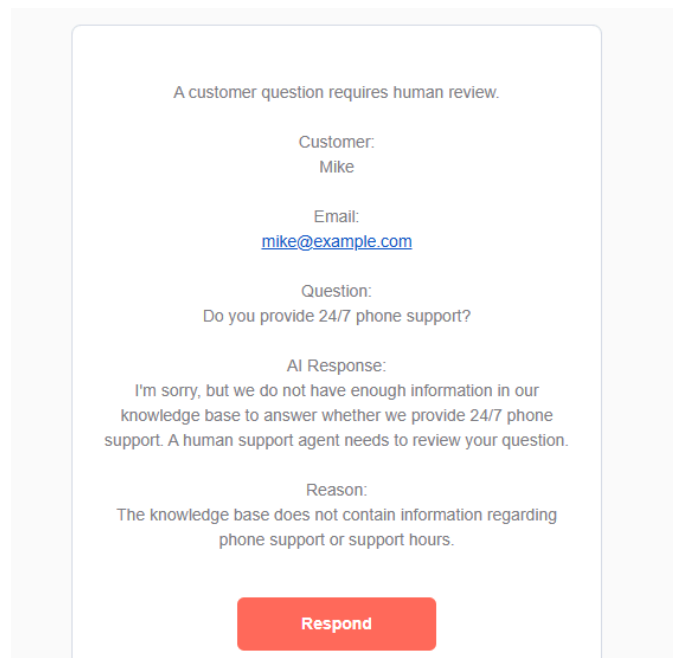
↓

Human Review

↓

Approved Response

This helps prevent the AI from confidently answering questions that need manual confirmation.



A customer question requires human review.

Customer:
Mike

Email:
mike@example.com

Question:
Do you provide 24/7 phone support?

AI Response:
I'm sorry, but we do not have enough information in our knowledge base to answer whether we provide 24/7 phone support. A human support agent needs to review your question.

Reason:
The knowledge base does not contain information regarding phone support or support hours.

Respond

Human review trigger — query is escalated when KB lacks required support information.

Knowledge Base & Vector Search

The system converts document chunks into embeddings using:

Gemini Embeddings — gemini-embedding-001

The embeddings are stored in a Supabase vector table and used for semantic retrieval.

This allows the system to find relevant information even when the user's wording does not exactly match the source document.

Parameters Settings

Credential
Google Gemini API Safin

Each model is using different dimensional density for embeddings. Please make sure to use the same dimensionality for your vector store. The default model is using 768-dimensional embeddings.

Model
models/gemini-embedding-001

OUTPUT

1 item

response

response[0]

# 0[0]	-0.008231191
# 0[1]	-0.00077404646
# 0[2]	0.011616955
# 0[3]	-0.078784175
# 0[4]	-0.0018906828
# 0[5]	0.009653598
# 0[6]	0.025512256
# 0[7]	0.0001800508
# 0[8]	-0.015994843
# 0[9]	-0.034418147
# 0[10]	-0.019222867
# 0[11]	-0.0044984263
# 0[12]	-0.024317687

Press the "Insert" on your keyboard to take a screenshot

Gemini Embeddings (gemini-embedding-001) used to vectorize knowledge-base chunks.

Expected Business Outcome

This is a portfolio project, not a real client engagement, so no real performance metrics are claimed.

Expected outcomes:

- Faster customer responses
- More consistent support answers
- Less repetitive support work
- Safer handling of uncertain questions
- Easier access to internal knowledge

Business Use Cases

This system can be adapted for:

- SaaS Support
Answer product and account questions.
- E-commerce
Answer questions about products, orders, and policies.
- Agencies
Create internal support assistants for teams and clients.
- Real Estate
Answer questions using property and company documentation.
- Internal Helpdesk
Search company policies, processes, and internal documents.

Key Takeaway

- This system connects a company's existing documentation to an AI support workflow.
 - Index → Retrieve → Answer → Escalate when needed
 - The result is a support assistant that can use internal knowledge while keeping a human involved when the information is not sufficient.
-

Tools & Technologies

Automation: n8n

AI: Google Gemini

Embeddings: Gemini Embeddings

Knowledge Source: Google Docs / Google Drive

Vector Database: Supabase

Retrieval: Retrieve Relevant Documents

Human Review: Gmail / Send & Wait

Output: Structured JSON